where we comment that α can be seen as a continuum approximation allowing us to replace data summation with integrals over the data measure and β is a combination of mean-field-type scaling Mei et al. (2018) together with a thermodynamic/saddle-point limit. Notably the following combination (u) of hyper-parameter $u = \frac{n^2\sigma_a^2}{\sigma^4 dN}$, which we refer to as the effective interaction, is invariant under both α and β.

Claim I. Two relevant discrepancy modes before the transition. For $\beta \to \infty$ and $\sqrt{\beta}/\alpha \to 0$, and $u \leq u_c(\epsilon)$ (≈ 30.2 for $\epsilon = -0.3$) the discrepancy takes the following form

$$\sigma^2 \bar{t}\left(\boldsymbol{x}\right) = bH_1\left(\boldsymbol{x}\right) + cH_3\left(\boldsymbol{x}\right) \tag{12}$$

where $b, c \in \mathbb{R}$ are some $O(1)$ constant coefficients. We comment that ub^2 is proportional to the emergent scale of Ref. Seroussi et al. (2023). For further detail see App. A.2.

Claim II. One-dimensional posterior weight distribution. In the same scaling limit, $u \leq u_c$ the negative log probability (action in physics terminology) of weights along the teacher direction, decouples from the rest of the $\boldsymbol{w}$ modes, and takes the following form

$$\mathcal{S}[\boldsymbol{w}\cdot\boldsymbol{w}^*] = d\left(\frac{\left(\boldsymbol{w}\cdot\boldsymbol{w}^*\right)^2}{2\sigma_w^2} - \frac{2n^2\sigma_a^2}{\pi\sigma^4 dN}\frac{\left(\boldsymbol{w}\cdot\boldsymbol{w}^*\right)^2}{1+2\left(\sigma_w^2+\left(\boldsymbol{w}\cdot\boldsymbol{w}^*\right)^2\right)}\left(b - \frac{2c\left(\boldsymbol{w}\cdot\boldsymbol{w}^*\right)^2}{1+2\left(\sigma_w^2+\left(\boldsymbol{w}\cdot\boldsymbol{w}^*\right)^2\right)}\right)^2\right) \tag{13}$$

Notably, this expression reduces the high-dimensional network posterior into a scalar probability involving only the relevant order parameter ($\Phi = \boldsymbol{w}\cdot\boldsymbol{w}^*$). We further note that all the expressions in the brackets are invariant under α and β. Heuristically, this will also be the action for a small Δu after the phase transition since the resulting corrections to the discrepancy have a parametrically small effect. For further detail see App. A.2.

Claim III. Exactness of Gaussian Mixture Approximation. In the same scaling limit, the probability described by the above action is exactly a mixture of Gaussians each centred around a global minimum of $\mathcal{S}$.

Claim IV. First-order phase transition. For $u < u_c$ the only saddle is that at $\Phi = 0$. Exactly at $u = u_c$ three saddles appear two of which are roughly at $\Phi = \pm|w_*|^2 = \pm 1$. For some finite interval $u \in [u_c, u_c + \Delta u]$, these saddles stay degenerate in $\hat{\mathcal{S}}$ value.

3.1.1 TEACHER-STUDENT EXPERIMENTAL RESULTS

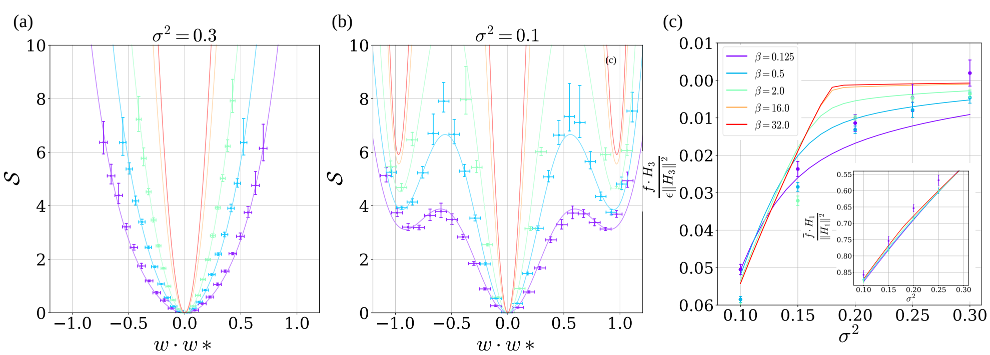

Figure 2: **GFL to GMFL-I Theory and Experiment in the teacher-student model.** Panel (a) and (b) show the negative log posterior before and after the phase transition induced by varying the noise σ^2. A good match with the experimental values (crosses) and theory (solid lines) for rarer and rarer events is obtained as we scale up the model according to Table 1. (colour coding). Turning to network outputs, panel (c) shows the expected phase transition in learning the cubic teacher component and the inset shows the discrepancy in the linear teacher component. As our analytics holds before and at the phase transitions, discrepancies in $\bar{f}\cdot H_3$ close to the transition are an expected finite-size effect made pronounced by its low absolute value.

To validate our theoretical approach, we trained an ensemble of 200 DNNs for different $\alpha = \beta$ values using our Langevin dynamics at sufficiently low learning rates and for a sufficiently long time to ensure equilibration. Our initial $N, d, \sigma^2, \sigma_a^2$ (i.e. at $\alpha = \beta = 1$) were: $N = 700, n = 3000, d = 150, \sigma_w^2 = 0.5, \sigma_a^2 = 8/N$ and we took $\epsilon = -0.3$. Our training ensemble consisted of both different initialization seeds and different data-draw seeds. These, together with the neuron indices associated N, provided $200N$ draws from $\boldsymbol{w}_* \cdot \boldsymbol{w}$ used for estimated $p(\boldsymbol{w})$. The discrepancy was estimated using a dot product of the outputs with $H_{1/3}(x)$ sampled over 3000 test points and then averaged over the 50 seeds.

Our experimental results are given in Fig. 2. Panel (c) shows how the linear (inset) and cubic target components learned, as a function of σ^2. Notably, reducing σ^2 is similar to increasing n, hence one expects more feature learning at lower σ^2. As we increase $\alpha = \beta$ (see color coding) a sharp phase transition develops around $\sigma^2 = 0.18$ where the cubic component begins its approach to the teacher's value (ϵ). Panels (a) and (b) track $-\log(p(\boldsymbol{w} \cdot \boldsymbol{w}_*))$ at two points before and after the phase transition, respectively. As $\alpha = \beta$ increases, the theory predicts a finite probability for $\Phi(\boldsymbol{w}) = \boldsymbol{w} \cdot \boldsymbol{w}_* \approx 1$. This means that picking a random neuron has a finite chance of fluctuating around the teacher-aware minima. Such neurons are what we refer to as our droplets. All graphs show a good agreement with theory as one scales up $\alpha = \beta$ before the transition. The remaining discrepancies are attributed to finite size (i.e. finite α, β) effects which, as expected, become more noticeable near the transition.

The above experiment also points to some potentially powerful complexity aspects of feature learning. Notably, an FCN NNGP kernel induces a uniform prior over cubic polynomials (i.e. all $l = 3$ hyper-spherical harmonics). As there are $n = O(d^3)$ of those, it requires $n = 0.5e + 6$ datapoints to learn such target components in our setting (see also Ref. Cohen et al. (2021)). Here this occurs two orders of magnitudes earlier ($n = 3000$). This occurs because the complex prior induced by a *finite* DNN learns the features from the readily accessible linear components of the target and applies them to the non-linear ones. Proving that such "assisted learning" of complex features changes the sample complexity compared to NNGPs requires further work. In App. A.1 we provide an analytical argument in support of this.

Table 1: Scaling laws

Model	Width	Input-dimension	Data size	Noise strength	Weight decay
Polynomial regression	$N \to \beta N$	$d \to \sqrt{\beta} d$	$n \to \alpha n$	$\sigma^2 \to \frac{\alpha}{\beta}\sigma^2$	$\sigma_a^2 \to \sigma_a^2/\sqrt{\beta}$
Modular Theory	$N \to \beta^2 N$	$P \to \sqrt{\beta} P$	$P^2 \to \beta P^2$	$\sigma^2 \to \sigma^2/\beta$	$\sigma_a^2 \to \sigma_a^2/\beta$

3.2 Modular Algebra Theory

Scaling setup and effective interaction.

Similar to the polynomial regression problem, we consider a scaling variable (β) which we later take to infinity together, and consider scaling up the microscopic parameters. The precise scaling is given in Table 1.

$$N \to \beta^2 N \quad P \to \sqrt{\beta} P \quad \sigma_a^2 \to \sigma_a^2/\beta \quad \sigma^2 \to \sigma^2/\beta \tag{14}$$

Note that, here, we do not need additional scale for the continuum limit of the dataset, since the continuum limit is taken by considering all combinations of data points. As before, β is a combination of mean-field scaling together with a thermodynamic/saddle-point limit. Notably, the following combination (u) of hyperparameter $u = \frac{2\sigma_a^2 P^2}{N\sigma^4}$, which we refer to as the effective interaction, is invariant under β.

Problem Symmetries

The following symmetries of $S[\boldsymbol{w}]$ (which is also a function of $\bar{\boldsymbol{t}}$) help us decouple the posterior probability distribution into its Fourier modes and simplify the problem considerably:

I. Taking $[n, m] \to [(n + q) \bmod P, (m + q') \bmod P]$, and $f_p \to f_{(p+q+q') \bmod P}$ with $q, q' \in \mathbb{Z}_P$

II. Taking $[n, m] \to [qn \bmod P, qm \bmod P]$ and $f_p \to f_{qp \bmod P}$ for $q \in \mathbb{Z}_P$ but different than zero.

Claim I. Single discrepancy mode. Several outcomes of these symmetries are shown in App. (B.1). First, we find that the adaptive GP kernel Q (given explicitly in Eq. 21) is diagonal in the basis $\phi_{k,k'}(x_{n,m}) = P^{-1}e^{2\pi i(kn+k'm)/P}$, where $k, k' \in \{0, 1, ..., P-1\}$. Considering eigenvalues, the second symmetry implies that $\phi_{k,k'}$ would be degenerate with $\phi_{ck,ck'}$. For prime, P this implies, in particular, that all $\phi_{k,k}$ eigenvectors with $k > 0$ have the same eigenvalue (λ). Notably, the target itself is spanned by this degenerate subspace specifically

$$y^p_{nm} = P^{-1}\sum_{k=1}^{P-1} e^{-i2\pi kp/P}e^{i2\pi k(n+m)/P} = \left[\sum_{k=1}^{P-1} e^{-i2\pi kp/P}\phi_{k,k}(x_{n,m})\right] \tag{15}$$

As a result, one finds that the target is always an eigenvector of the kernel and $\sigma^2\bar{t}^p_{mn} = ay^p_{nm}$ where $a \in \mathbb{R}$. Thus there is only one mode of the discrepancy which is aligned with the target.

Claim II. Decoupled two-dimensional posterior weight distribution for each Fourier mode. We decouple all the different fluctuating modes by utilizing again the symmetries of the problem and making a judicial choice of the non-linear weight decay term (Γ). To this end, we define the following Fourier transformed weight variables (w_k, v_k): $w_n = \sum_{k=0}^{P-1} w_k e^{-\frac{2\pi ikn}{P}}, \quad w_m = \sum_{k=0}^{P-1} v_k e^{-\frac{2\pi ikm}{P}}$ which when placed into action yields

$$\mathcal{S}[\hat{\boldsymbol{w}}] = P\left[\left(\frac{1}{2}\sum_{k=0}^{P-1} w_k w_{-k} + \frac{1}{2}\sum_{k=0}^{P-1} v_k v_{-k}\right) - \frac{2\sigma_a^2 a^2 P^2}{N\sigma^4}\sum_{k=1}^{P-1} w_k w_{-k} v_k v_{-k}\right] + \Gamma[\boldsymbol{w}] \tag{16}$$

(see App. B.2) where apart from the non-linear weight-decay term, all different k modes have been decoupled. For simplicity, we next choose, $\Gamma[\boldsymbol{w}] = \sum_k P\frac{\gamma}{6}\left((w_k w_{-k})^3 + (v_k v_{-k})^3\right)$. Here there are technically two order parameters- $\Phi = w_k w_{-k}, \Psi = v_k v_{-k}$, but from the symmetry of the action, we obtain that the saddles occur only at points where $\Phi = \Psi$. We comment that the analysis of a more natural weight decay terms such as $\sum_n w_n^6 + w_m^6$, using a certain GP mixture ansatz of $p(w_i)$ as an approximation, we obtained similar qualitative results.

Claim III. Exactness of Gaussian Mixture Approximation. Following the presence of a large factor of P in front of the action, the above non-linear action, per $k-$mode, can be analyzed using standard saddle point treatment. Namely, treating Q as a function of a, and with it λ, can be evaluated through a saddle-point approximate on the probability associated with this action. In the limit of $\beta \to \infty$, we obtain that this approximation is exact. Following, this a can be calculated using the GPR expression $-\frac{\sigma^2}{\lambda+\sigma^2}$, and in this limit the value of λ can be computed exactly. Demanding this latter value of a matches the one in the action results in an equation for a.

Claim IV. First-order phase transition. Since the quadratic term is constant in the scaled action, as long as no other saddles become degenerate (in action value) with the $\Phi = \Psi = 0$ saddle, the saddle-point treatment truncates the action at this first term. By increasing u the quartic term becomes more negative, and hence a first-order symmetry-breaking transition must occur at some critical value of u. Past this point, a begins to diminish. If it will diminish too rapidly, the feedback on the action will be such that it is no longer preferential for a to diminish, and thus a probability distribution will have two degenerate minima at a zero and non zero value of Φ representing the GFML-I phase. Further increasing u will break the degeneracy resulting in the global minimum of the log posterior distribution being non trivial. Notably a measures the test-RMSE here, thus the fact that it remains constant and suddenly begins to diminish can also be understood as Grokking.

3.2.1 Modular Algebra Numerical Simulations

Solving the implied equation for a numerically yields the full phase diagram here (see App. B.3 for technical details of solution) supporting the assumptions in **Claim IV**. Fig. (3) plots the negative-log-probability of weights for an arbitrary k taking $\Phi = \Psi$ for simplicity, since at the global minima this is anyways true. Here we increase u by decreasing σ^2 and find the action by solving the equation for $a(\sigma^2)$ numerically. The $\Phi \neq 0$ saddles are exponentially suppressed in P, yet nonetheless become more probable as σ^2 decreases. At around, $\sigma^2 \approx 0.227$ they come within $\mathcal{O}(1)$ of the saddle at $S_k(\Phi = 0, \Psi = 0)$ ($\mathcal{O}(1/P)$ in the plot, given the scaled y-axis). This marks the beginning of the mixed phase (GMFL-I), wherein all action minima contribute in a non-negligible manner. Further, in this phase, decreasing σ^2 does not change the height of the $\Phi \neq 0$ minima (see inset) in any appreciable manner. Had we zoomed in further, we would see a very minor change to these saddle's height throughout the mixed phase, as they go from being $\mathcal{O}(1)$ above the $\Phi = 0$ saddle to $\mathcal{O}(1)$ below that saddle at $\sigma^2 \approx 0.175$, this is shown in the inset graph in Fig. (3). This latter point marks the beginning of the GMFL-II phase, where it is the contribution of the minimum at $\Phi = 0$ which becomes exponentially suppressed in P. Notably a, which measures here the test-RSME, goes from -1 at the beginning of GMFL-I to -0.7 at its end. Over this small interval of σ^2 we observe a 30% reduction in the magnitude of the discrepancy which can be thought of as a manifestation of Grokking.

Our analytical results are consistent with the experiments carried out in Ref. Gromov (2023). Indeed, as we enter GMFL-I, weights sampled near the $\Phi, \Psi \neq 0$ saddle, correspond to the cosine expressions for $W_k^{(1)}$ of that work. As our formalism marginalizes over readout layer weights, the phase constraint suggested in their Eq. (12) becomes irrelevant, and both viewpoints retrieve the freedom of choosing cosine phases. In our case, this stems from the $U(1) \times U(1)$ complex-phase freedom in our choice of saddles at $\Phi, \Psi > 0$.

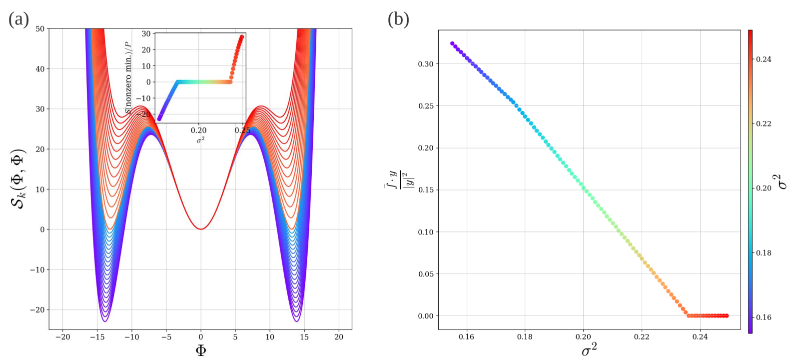

Figure 3: **GFL to GMFL-I to GMFL-II** (a) Probability distribution of weights as predicted by our approach. The GFL phase is represented by the red graphs, where the minimum of the action at zero (shared also by the GP limit) dominates the probability distribution. The GMFL-I phase can be seen in the inset graph, and the final GMFL-II phase is shown in purple.(b) The target component of the average network output. Here singularities can be observed at the σ^2 values where the phase transitions occur. The Parameters taken in this calculation are: $N = 1000, P = 401, \sigma_a^2 = 0.002/N, \gamma = 0.0001$

4 Discussion

In this work, we studied two different models exhibiting forms of Grokking and representation/feature learning. We argue that Grokking is a result of a first order phase transition. To analyze this analytically, we extended the approach of Seroussi et al. (2023) to include mixtures of Gaussian. The resulting framework led to concrete analytical predictions for rich feature learning effects exposing, in particular, several phases of learning (GFL,GMFL-I,GMFL-II) in the thermodynamic/large-scale limit. Our results also suggest that feature learning in finite FCNs with mean-field scaling can change the sample complexity compared to the associated NNGP. Certainly, these describe very different behavior compared to the recently explored kernel-scaling Li & Sompolinsky (2021); Ariosto et al. (2022) approach, wherein feature learning amounts to a multiplicative factor in front of the output kernel. A potential source of difference here is their use of standard scaling, however, this remains to be explored.

As our results utilize a rather general formalism Seroussi et al. (2023), we believe they generalize to deep networks and varying architecture. As such, they invite further examination of feature learning in the wild from the prism of the latent kernel adaptation . Such efforts may provide, for instance, potential measures of when a model is close to Grokking by tracking outliers in the weight or pre-activation distributions along dominant kernel eigenvectors. As latent kernels essentially provide a spectral decomposition of neuron variance, they may help place empirical observations on neuron sensitivity and interpretability Zeiler & Fergus (2014) on firmer analytical grounds. Finally, they suggest novel ways of pruning and regulating networks by removing low-lying latent kernel eigenvalues from internal representations.

REFERENCES

S Ariosto, R Pacelli, M Pastore, F Ginelli, M Gherardi, and P Rotondo. Statistical mechanics of deep learning beyond the infinite-width limit. *arXiv preprint arXiv:2209.04882*, 2022.

Luca Arnaboldi, Ludovic Stephan, Florent Krzakala, and Bruno Loureiro. From high-dimensional & mean-field dynamics to dimensionless ODEs: A unifying approach to SGD in two-layers networks. *arXiv e-prints*, art. arXiv:2302.05882, February 2023. doi: 10.48550/arXiv.2302.05882.

Gerard Ben Arous, Reza Gheissari, and Aukosh Jagannath. Online stochastic gradient descent on non-convex losses from high-dimensional inference. *The Journal of Machine Learning Research*, 22(1):4788–4838, 2021.

Alberto Bietti, Joan Bruna, Clayton Sanford, and Min Jae Song. Learning single-index models with shallow neural networks. *Advances in Neural Information Processing Systems*, 35:9768–9783, 2022.

Blake Bordelon and Cengiz Pehlevan. Self-consistent dynamical field theory of kernel evolution in wide neural networks. *Advances in Neural Information Processing Systems*, 35:32240–32256, 2022.

Supriyo Chakraborty, Richard Tomsett, Ramya Raghavendra, Daniel Harborne, Moustafa Alzantot, Federico Cerutti, Mani Srivastava, Alun Preece, Simon Julier, Raghuveer M. Rao, Troy D. Kelley, Dave Braines, Murat Sensoy, Christopher J. Willis, and Prudhvi Gurram. Interpretability of deep learning models: A survey of results. In *2017 IEEE SmartWorld, Ubiquitous Intelligence & Computing, Advanced & Trusted Computed, Scalable Computing & Communications, Cloud & Big Data Computing, Internet of People and Smart City Innovation (SmartWorld/SCALCOM/UIC/ATC/CBDCom/IOP/SCI)*, pp. 1–6, 2017. doi: 10.1109/UIC-ATC.2017.8397411.

Omry Cohen, Or Malka, and Zohar Ringel. Learning curves for overparametrized deep neural networks: A field theory perspective. *Physical Review Research*, 3(2):023034, 2021.

Alain Durmus and Eric Moulines. Nonasymptotic convergence analysis for the unadjusted langevin algorithm. *The Annals of Applied Probability*, 27(3):1551–1587, 2017.

David Gamarnik, Eren C Kızıldağ, and Ilias Zadik. Stationary points of shallow neural networks with quadratic activation function. *arXiv preprint arXiv:1912.01599*, 2019.

E. Gardner and B. Derrida. Optimal storage properties of neural network models. *Journal of Physics A Mathematical General*, 21:271–284, January 1988. doi: 10.1088/0305-4470/21/1/031.

Andrey Gromov. Grokking modular arithmetic. *arXiv preprint arXiv:2301.02679*, 2023.

Géza Györgyi. First-order transition to perfect generalization in a neural network with binary synapses. *Phys. Rev. A*, 41:7097–7100, Jun 1990. doi: 10.1103/PhysRevA.41.7097. URL `https://link.aps.org/doi/10.1103/PhysRevA.41.7097`.

Lev Davidovich Landau and Evgenii Mikhailovich Lifshitz. *Statistical Physics: Volume 5*, volume 5. Elsevier, 2013.

Qianyi Li and Haim Sompolinsky. Statistical mechanics of deep linear neural networks: The backpropagating kernel renormalization. *Phys. Rev. X*, 11:031059, Sep 2021. doi: 10.1103/PhysRevX.11.031059. URL `https://link.aps.org/doi/10.1103/PhysRevX.11.031059`.

Ziming Liu, Ouail Kitouni, Niklas Nolte, Eric J Michaud, Max Tegmark, and Mike Williams. Towards understanding grokking: An effective theory of representation learning. In Alice H. Oh, Alekh Agarwal, Danielle Belgrave, and Kyunghyun Cho (eds.), *Advances in Neural Information Processing Systems*, 2022a. URL `https://openreview.net/forum?id=6at6rB3IZm`.

Ziming Liu, Eric J Michaud, and Max Tegmark. Omnigrok: Grokking beyond algorithmic data. *arXiv preprint arXiv:2210.01117*, 2022b.